COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 6

3 Complexity Theory (ad260)

Recall that, for languages A and B, $A \leq_P B$ denotes that A is *polynomial-time reducible* to B.

Reductions

(*a*) Show that if $A \leq_P B$ and $B \leq_P C$, then $A \leq_P C$. [3 marks]

Answer: Let f be the reduction from A to B and g the reduction from B to C. That is, f and g are polynomial-time computable functions such that for any x, $x \in A$ iff $f(x) \in B$ and $x \in B$ iff $g(x) \in C$. Then, $g \circ f$ is a polynomial-time reduction from A to C. It is immediate that $x \in A$ iff $g(f(x)) \in C$. To set that $g \circ f$ is computable in polyomial time, note that the length of $f(x)$ is bounded by a polynomial in the length of x.

reductions and completeness

(*b*) Show that if A is NP-complete and $A \leq_P B$ and $B \leq_P A$, then B is also NP-complete. [3 marks]

Answer: Since A is NP-complete, for every language L in NP we have $L \leq_P A$. Then, since $A \leq_P B$, we have, by part (a), $L \leq_P B$. Thus, B is NP-hard and it remains to argue that B is in NP. But, A is in NP, and $B \leq_P A$. It follows that B is in NP, since this class is closed under polynomial-time reductions.

For any $k \in \mathbb{N}$, let k-EC (*exact cover by k sets*) denote the language coding the following decision problem.

> Given a set X and a collection S of k-element subsets of X, determine if there exists a set $C \subseteq S$ such that each $x \in X$ appears in exactly one element of C.

Matching, set-covering

(*c*) Show that, for each fixed k, k-EC $\leq_P$ $(k+1)$-EC. [5 marks]

Answer: We describe a reduction from k-EC to $(k+1)$-EC. First of all, note that for an instance (X, S) to admit an exact cover by k sets the size of X must be a multiple of k.

So, if the size of X is not a multiple of k, return any fixed instance (X', S') of $(k+1)$-EC where the size of X' is not a multiple of $k+1$.

If the size of X is mk for some m, we define X' to be X extended by exactly m new elements $x_1, \ldots, x_m$. We define S' to be the set of $k+1$-element subsets of X' obtained from some set $a \in S$ by adding to it exactly one of the new elements. The output is (X', S').

This function is clearly computable in polynomial time, so it remains to argue that (X, S) admits an exact cover by k sets if, and only if, (X', S') admits an exact cover by $k+1$-sets. So, suppose (X, S) admits such a cover $C \subseteq S$. Since X has exactly km elements, there are exactly m sets in C. For each $a \in C$ choose a distinct element x_i from among $x_1, \ldots, x_m$ and note that $a \cup \{x_i\}$ is in S'. The resulting collection of m sets is an exact cover of X'. Conversely, suppose that (X', S') admits an cover C'. Then, C' has exactly m sets in it. Moreover, since no pair of elements from among $x_1, \ldots, x_m$ appears together in any set in S', each set in C' must contain exactly one of them. Removing them then gives an exact cover in (X, S).

For each of the following statements, determine whether the it is true, false or unknown and give full justification for your answer. If the answer is unknown, the justification may be that it is equivalent to some well-known open problem. You may assume standard facts about the complexity of matching and covering problems covered in the lectures, as long as you state them clearly.

(*d*) 2-EC $\leq_P$ 4-EC [3 marks]

reductions

Answer: This is true, immediately by parts (a) and (c).

(*e*) 4-EC $\leq_P$ 3-EC [3 marks]

NP-completeness, set covering

Answer: This is true because 3-EC is NP-complete (a fact proved in the lectures) and 4-EC is in NP.

(*f*) 4-EC $\leq_P$ 2-EC [3 marks]

polynomial-time, matching

Answer: This is unknown. It is equivalent to the statement P=NP. Indeed, 2-EC is solvable in polynomial time. It is the same as the general graph matching problem. 4-EC is NP-complete by the fact that 3-EC is NP-complete, together with (c) and (b).